

JOSHUA MUHUMUZA

AI Safety Researcher | Senior Systems Engineer | AI Governance Practitioner

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PROFILE

AI safety researcher and systems engineer with demonstrated impact across clinical AI deployment, mechanistic interpretability, and AI governance. A Senior systems engineer with over 7 years in designing and developing Enterprise systems that have over 10,000+ users worldwide.

EDUCATION

MSc Computer Science (AI Track) — Makerere University 2022 – 2024
GPA: 4.66/5 - First Class Honors | Sponsored by Google DeepMind

BSc Software Engineering — Makerere University 2014 – 2018
GPA: 4.78/5 — First Class Honours | Sponsored by Government of Uganda

RESEARCH PROJECTS

AI Safety: Mechanistic Interpretability in Clinical AI 2026 – Present

- Identified a spurious attention circuit (L8H10) in a Vision Transformer trained on chest X-rays that is causally verified as harmful yet completely invisible to standard output-level safety evaluation, with direct implications for clinical AI procurement and governance in low-resource settings.
- Finding replicates across architectures and datasets. Manuscript submitted to Deep Learning Indaba 2026 / IJCAI.

Pluralistic Alignment: Annotation Bias & Hate Speech 2025 – Present

- Demonstrated empirically that majority-vote annotation in hate speech datasets suppresses minority value judgments at the hate/offensive boundary, a structural failure with direct implications for multilingual, multicultural populations across East Africa.
- Paper Accepted and published at ICML conference under Pluralistic Alignment Workshop @ ICML 2026.

ITU Project: University of Oxford February 2025 – May 2025

- Collaborated on country-focused policy case studies (primarily Kenya), analyzing national responses to emerging AI safety institutes and governance structures. (Funded by Institute of Applied Policy)

AMR Surveillance AI Model October 2024 – Present

- Served as Tech Lead in designing and piloting a low-resource recommendation models and companion mobile application for antimicrobial resistance (AMR) surveillance, co-infection detection, and real-time information dissemination to community. (Funded By: GIZ).
- Engineered efficient models optimized for low-compute environments to enable accurate monitoring and identification of AMR threats in data.

Sickle Cell Early Detection AI June 2022 – Present

- Led development of a lightweight deep learning model for enhanced, accurate interpretation of rapid diagnostic tests in low-resource contexts, improving early sickle cell disease detection. (Funded By: National Institutes of Health grant).
- Optimized computer vision algorithms to boost diagnostic reliability on mobile devices, addressing accessibility challenges in underserved regions.

Tuberculosis Classifier and report Generation AI Jan 2023 – Sept 2023

- Directed creation of an explainable multi-task deep learning architecture for chest X-ray classification and automated report generation, overcoming long-context limitations in large language models for clinical usability. (Funded by Lacuna Fund)
- Built an interpretable AI system that streamlines TB screening workflows, with potential to accelerate diagnosis in high-burden settings.

GRANTS & FUNDING

RUFORUM Grant: Integrated Learning Management System \$160,000 | 2026
Regional Universities Forum for Capacity Building in Agriculture (RUFORUM)

- Awarded to design and deploy an integrated learning management system serving 175+ universities across Africa, strengthening shared academic infrastructure at continental scale.

Buganda Royal Institute Grant: Vocational Academic Management System \$20,000 | 2026
Buganda Royal Institute Uganda

- Awarded to develop an academic management system for vocational training institutions, supporting skills development and economic participation across the region.

NIH Grant: Sickle Cell Early Detection 2023 – Present
National Institutes of Health USA

- Co-investigator on NIH-funded project developing lightweight sickle cell detection models for mobile deployment in low-resource clinical settings in Uganda.

AMR Surveillance Grant: ADAPT One Health

2024 – Present

BFTR and GIZ Germany

- Funded development and deployment of an AMR Mobile Application and dashboard to support the National AMR board in tracking AMR and promote education awareness in Uganda.

Weather & Climate Grant: WIMEA

2024 – Present

NORHED

- Funded development and deployment of design and monitoring of automatic weather stations in East Africa.

Lacuna Fund Grant: TB Chest X-Ray Classification

2022 – Present

Lacuna Fund

Remote

- Funded development and deployment of multi-task deep learning model for TB classification and report generation at Uganda's national referral hospitals.

WORK EXPERIENCE

Senior Systems Engineer

Jan 2021 – Present

Makerere University

Kampala, Uganda

- Built Makerere University Data Warehouse (MakData), first scalable integration of siloed institutional systems, processing 25TB+ of data and serving 175+ universities across Africa.
- Developed the Academic Records System (MakARS), digitising records from the 1900s and managing 15TB+ of institutional data under the DARP project.
- Developed Graduate e-Supervision System (RIMS), contributing to a 60% increase in on-time graduations; adopted in CAES with pilot expansions to CHUSS and CEES.
- Designed and deployed Mak e-Vote, the university's secure electronic voting platform; managed five consecutive guild elections as appointed oversight committee member.
- Developed Electronic Human Resource Management System (e-HRMS), automating the full HR lifecycle across the institution.

Google DeepMind Scholar

Jan 2022 – Sept 2024

Google DeepMind

London, UK

- Designed and fine-tuned ML models for health diagnostics in Uganda; two projects secured NIH and Lacuna Fund grants and were deployed at national referral hospitals.

Lead AI Developer

Dec 2018 - Dec 2020

Emaisha Ltd

Kampala, Uganda

- Led team that designed and launched Uganda's first farmer call center, securing the company's inaugural UNCDF grant and expanding agricultural support services.
- Developed Livestock Manager web and Android portal for efficient livestock management.
- Engineered a USSD-based farm management application for low-connectivity areas, promoting digital inclusion and service delivery to underserved farmers.

Software Engineer

Aug 2018 - Dec 2018

Thinvoid Ltd

Kampala, Uganda

- Designed and prototyped data management system for NGO Educate!, boosting productivity across 700+ schools in Africa.

Software Developer

Jun 2017 - Jun 2018

WIMEA-ICT

Makerere University, Uganda

- Developed web portal for managing automatic weather stations; design adopted by Uganda National Meteorological Association.

PUBLICATIONS & PRESENTATIONS

Muhumuza, J. et al. (2025). **Spurious Circuits Invisible to Behavioural Evaluation: Mechanistic Interpretability in Clinical Vision Transformers.** *Submitted under review, Deep Learning Indaba 2026 / IJCAI Journal*

Muhumuza, J. et al. (2026). **When Majority Vote Silences Minority Values: Pluralistic Alignment Failures in Hate Speech Detection.** *Accepted, Pluralistic Alignment Workshop @ ICML 2026.*

Muhumuza, J., Katumba, A., Nakatumba-Nabende, et al. (2024). **A Multi-Task Deep Learning Model for Classification and Report Generation from Chest X-Ray Images.** *Under review.*

Nsabagwa M., Muhumuza J. et al. (2018). **Minimal Idle Listening and Centralized Scheduling in TSCH Wireless Sensor Networks.** *IEEE, Greece.* [doi](#)

Nsabagwa M., Muhumuza J. et al. (2018). **Condition Monitoring and Reporting Framework for WSN-based Automatic Weather Stations.** *IEEE, Botswana.* [doi](#)

AI GOVERNANCE & POLICY

Researcher: Kenya AI Safety Institute Framework

Feb – May 2025

ITU / Oxford University (Institute for AI Policy and Strategy)

Remote

- Researched the draft policy framework for Kenya's AI Safety Institute, one of the first such institutional frameworks produced for an African context, informing national-level AI governance design.
- Conducted cross-country analysis of AI safety governance structures across EAC partner states, documenting the near-total absence of safety evaluation infrastructure and the institutional risks this poses for deployed AI systems.

AI Strategy Contributor & Digital Leaders Forum Member

2026 – Present

East African Community / EASTECO

East Africa

- Active contributor to the EAC's regional AI strategy; participated in the 4th STI AI Conference on AI4EAC alongside policymakers, regulators, and civil society leaders from all EAC partner states, with GIZ as a key international partner.
- Advising on AI safety, equitable deployment, and governance frameworks for a region where AI adoption is accelerating without corresponding safety infrastructure or regulatory capacity.